

A Personal History of the Development of Immersive Video

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ABSTRACT

Immersive video, also known as 360 video, is a new creative medium that began in the 1990's. Here a pioneer gives his story of the origin of photographic VR with the making of the first movie in 1995, and the construction of the first recording systems, leading to the breakthrough development of the Dodeca 2360 production system, used for the city recordings employed by Google to launch Street View in 2007.

Keywords: Immersive Media, McCutchen, Dodeca.

Index Terms: Video virtual reality -- photographic virtual reality -- non-computer VR imaging -- early panoramic imaging -- early immersive development

1 THE BEGINNING

You might say that the story began in 1989 with my winning on *Jeopardy!* which enabled me to file the first of my patents for immersive video, entitled "Method and Apparatus for Dodecahedral Imaging System." I had been working in special effects in Hollywood, and was convinced that the development of new miniature video sensors would lead to a new form of wrap-around full motion pictures. The patent was focused on the most optically efficient and high-resolution approach for making a spherical image, using modular components oriented to a dodecahedron.

After I moved to Portland, Oregon, I founded Dodeca LLC in 1994 along with my friends Peter Giambrone and Stuart Ellis and got down to the work of building a working system after we got seed money from the cellular investor David Pardue. To get a sense of what immersive video might be like, I built a geodesic projection screen in the studio, which was on the second floor of Linnea Hall, an historic Victorian building that used to be a Swedish social club.

To make the multichannel video to fill the screen, I built the first immersive camera in 1994 with black and white board cameras on a dodecahedral frame. Video in those days was analog, and the only way to record it was on tape, with clusters of camcorders used as VCRs. The miniaturization of the cameras

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Figure 1 Projection dome inside office.

and recorders enabled the bundled system to be easily portable.



Figure 2 Projected four-channel immersive image

Projecting the video segments back on the screen confirmed that the wide-angle image would appear



Figure 3 David McCutchen and first B&W immersive camera 1994

natural and give you a convincing sense of being in a place. Building a new camera using small color cameras was the next step, and by the end of the year, the first prototypes were underway.

In Seattle, Virtual I/O was in the process of making the iGlasses, one of the first commercially available modern head mounted displays. We visited them late in 1994 to talk about making immersive photographic content to see in them, in addition to the usual computer-generated games. When I heard about the company Warp California who was showing a viewing program which could look around inside of a spherical image, I went to Las Vegas in January 1995 to find Eric Gullichsen at the Consumer Electronics Show, and we agreed to work together to make an immersive video movie for the Virtual I/O HMD exhibit at the top show for digital imaging, SIGGRAPH 95, which would be taking place in Los Angeles in August.

This would be the world's first immersive video movie.

2 The First Movie

The movie, entitled "The Sphere," was shot on April 16, 1995 in Portland, Oregon. The color camera used a patented flash synchronization system to synchronize the images, which were recorded by a bank of camcorders used as VCRs. A binaural head microphone system was used for maximum-quality spatial audio. The director for the video was Stuart Ellis, and Peter Giambrone was the producer.



Figure 4 David McCutchen and Stuart Ellis at recording for *The Sphere*

The equipment was arranged to be invisible to the camera. The basketball action took place around it, and the story was that a little kid and another guy take on two other players, and beat them, because the kid is actually a trick dribbler, and his partner is



Figure 5 The Sphere - first frame, full sphere view (remastered by David McCutchen 1999)

the city's slam dunk champion. It's fun, and following the players as they run around you is a good way to show off the HMD's head tracking.



Figure 6 The Sphere - middle and end

The recorded analog video was digitized first to make the 48-second movie. Because this was before the JPEG standard was in use, the files were relatively uncompressed BMP bitmaps and very large, which were a challenge to store and process with the slow processors available. Eric Gullichsen and Sue Wyshinsky of Warp California did the work aboard their converted ferry houseboat in the harbor in Sausalito, California, where Eric could sometimes be seen driving his Shelby. I remember seeing signs advertising other computer-generated VR projects being done at the time, which featured blocky shapes, rather than anything photographic.

3 THE PREMIERE

The premiere of the movie at SIGGRAPH '95 took place in Los Angeles on August 6, 1995, at the Virtual I/O booth, where people first had the first



Figure 7 Peter Giambrone assists a viewer. The HMD image can be seen on the monitor (photos by David McCutchen)



Figure 8 Stuart Ellis at the Virtual I/O booth

ever experience of being inside a photo-immersive movie.

The premiere was a success, with many surprised and awed reactions, especially when the movie was stopped by hitting the space bar, and people could look around within the still spherical frame. This was proof that this was a new kind of movie, where there was so much more to the image than what could just be seen in the frame.

SIGGRAPH (from Special Interest Group on Computer Graphics and Interactive Techniques) at that time was the most important convention and showcase for advancements in computer imaging. This was also the year when Apple introduced Eric Chen's program Quicktime VR, which included not only a viewer for looking outward within a spherical image, but also a viewer for looking inward, while moving around an object.



Figure 9 Enjoying the premiere at SIGGRAPH '95



Figure 10 Sue Wyshinsky (Warp), Eric Chen (Apple), and Eric Gullichsen (Warp California) (photo by David McCutchen)

4 The Aftermath

After the show, Warp California moved on to other projects, and the idea of immersive video was slow to catch on. It seemed that people were seeing immersive video as just a novelty, without commercial application. For the HMD, it was a choice between a game that you could play for a long time, and an immersive show that was over quickly. There was no backlog of immersive titles to choose from. The analog video hardware was still evolving. Computers were still slow, bandwidth was very narrow, and the Web had just begun.

And there was the confusion of the new. When I first started to use the word “immersive,” people assumed I meant “underwater.” This new medium challenged so many assumptions, even such fundamental things as the idea of framing a shot. So much about it was coming from a completely different direction.

On the bleeding edge.

These were lean years. Canadian investors led by David Anderson and later Myles McGovern kept us afloat, and the hardware development continued.



Figure 5 The engraved copper “eye” logo referred to the instinctive all-around awareness that immersive movies represented.

After the company name changed to the Immersive Media Company (IMC) in 1996, I built the improved Dodeca 1000 immersive camera at the camera manufacturer Cohu in San Diego. This new camera was still analog NTSC, with 640x480 VGA equivalent sensors, the best available, but featured

synchronized frames for all the channels, and unified digital controls for color and exposure. The camera control box case and the 8mm analog tape camcorders were mounted on a wheelchair for mobility, while keeping the cameraman and all the components, including lights, invisible to the camera.



Figure 12 David McCutchen with mobile recording system 1996

5 CONCLUSION

From these early days, the immersive movies multiplied as IMC’s hardware and software kept improving. I designed the Dodeca 2360 system to be easily portable and able to record for hours, with an automated postproduction process that enabled large projects. IMC’s full-motion, georeferenced, four-channel sound recordings of cities were used by Google to launch Street View in 2007.



Figure 13 IMC Camera car 2006 with Dodeca 2360 camera